Lab No: 2

SIMPLE AND MULTIPLE LINEAR REGRESSION

Task 1:

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_csv('diabetes_prediction_dataset.csv')

feature = 'bmi'
target = 'diabetes'

plt.figure(figsize=(8, 5))
plt.scatter(df[feature], df[target], alpha=0.5)
plt.title('BMI vs Diabetes Outcome')
plt.xlabel('BMI')
plt.ylabel('Diabetes (0 = No, 1 = Yes)')
plt.show()
```

```
correlation = df[feature].corr(df[target])
print(f"Correlation between BMI and Diabetes: {correlation:.3f}")

X = df[[feature]]
y = df[target]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

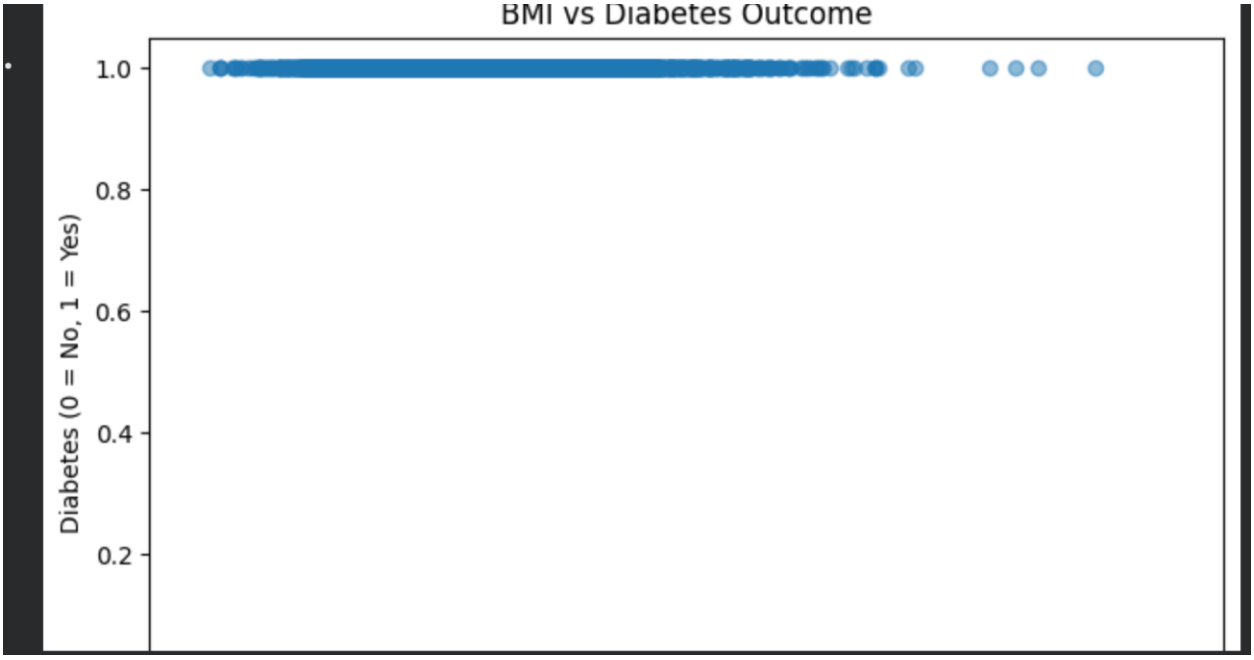
plt.figure(figsize=(8, 5))
plt.scatter(X_test, y_test, color='blue', alpha=0.5, label='Actual Data')
plt.plot(X_test, y_pred, color='red', linewidth=2, label='Regression Line')
plt.title('Simple Linear Regression: BMI vs Diabetes')
plt.legend()
plt.show()
```

```
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

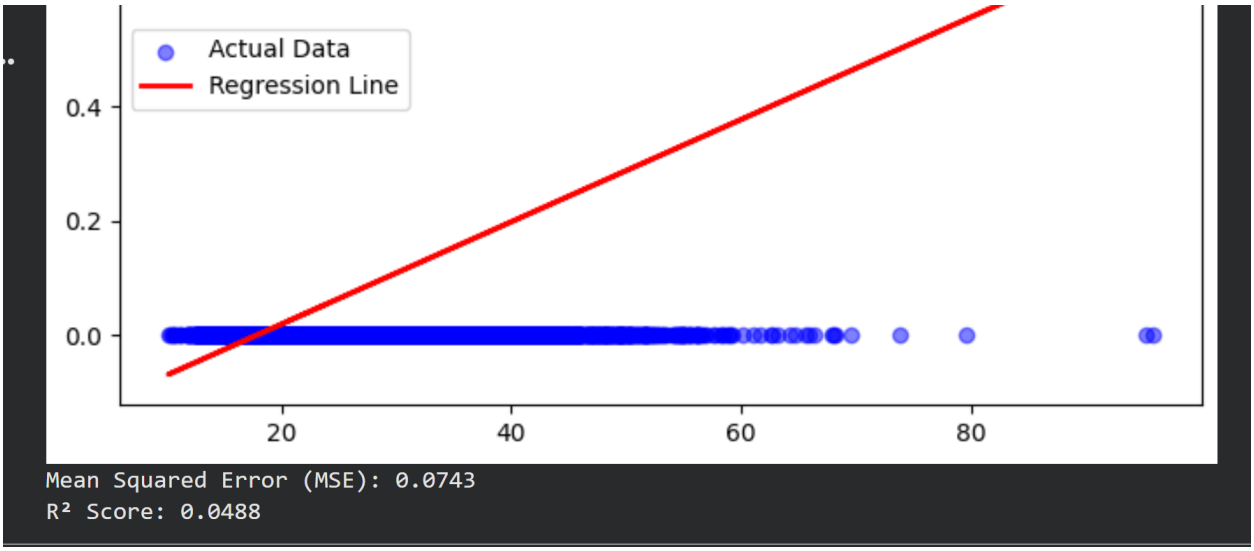
print(f"Mean Squared Error (MSE): {mse:.4f}")
print(f"R² Score: {r2:.4f}")
```



Output:


BMI vs Diabetes Outcome
Diabetes (0 = No, 1 = Yes)
1.0
0.8
0.6
0.4
0.2


BMI
Correlation between BMI and Diabetes: 0.214
Simple Linear Regression: BMI vs Diabetes
1.0
0.8
0.6
0.4
Actual Data
Regression Line



Answer:

No, BMI alone does not explain most of the variation in diabetes outcome. The R^2 score is 0.0488, which means BMI only explains about 4.88% of the variance in the target variable. This is a very low R^2, indicating that the other 95% of the variation is due to factors not included in this simple model.

Task 2:

Code:

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np

df = pd.read_csv('diabetes_prediction_dataset.csv')

df_encoded = pd.get_dummies(df, drop_first=True)

plt.figure(figsize=(12, 8))
sns.heatmap(df_encoded.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap')
```

```python
plt.figure(figsize=(12, 8))
sns.heatmap(df_encoded.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap')
plt.show()

X = df_encoded.drop('diabetes', axis=1)
y = df_encoded['diabetes']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model_multi = LinearRegression()
model_multi.fit(X_train, y_train)

y_pred_multi = model_multi.predict(X_test)

plt.figure(figsize=(8, 5))
plt.scatter(y_test, y_pred_multi, alpha=0.5)
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--', lw=2)
plt.xlabel('Actual')
```

```
plt.figure(figsize=(8, 5))
plt.scatter(y_test, y_pred_multi, alpha=0.5)
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--', lw=2)
plt.xlabel('Actual')
plt.ylabel('Predicted')
plt.title('Actual vs Predicted (Multivariate)')
plt.show()

residuals = y_test - y_pred_multi
plt.figure(figsize=(8, 5))
plt.scatter(y_pred_multi, residuals, alpha=0.5)
plt.axhline(y=0, color='r', linestyle='--')
plt.xlabel('Predicted Values')
plt.ylabel('Residuals')
plt.title('Residual Plot')
plt.show()

mse_multi = mean_squared_error(y_test, y_pred_multi)
rmse_multi = np.sqrt(mse_multi)
r2_multi = r2_score(y_test, y_pred_multi)
```

```
plt.show()

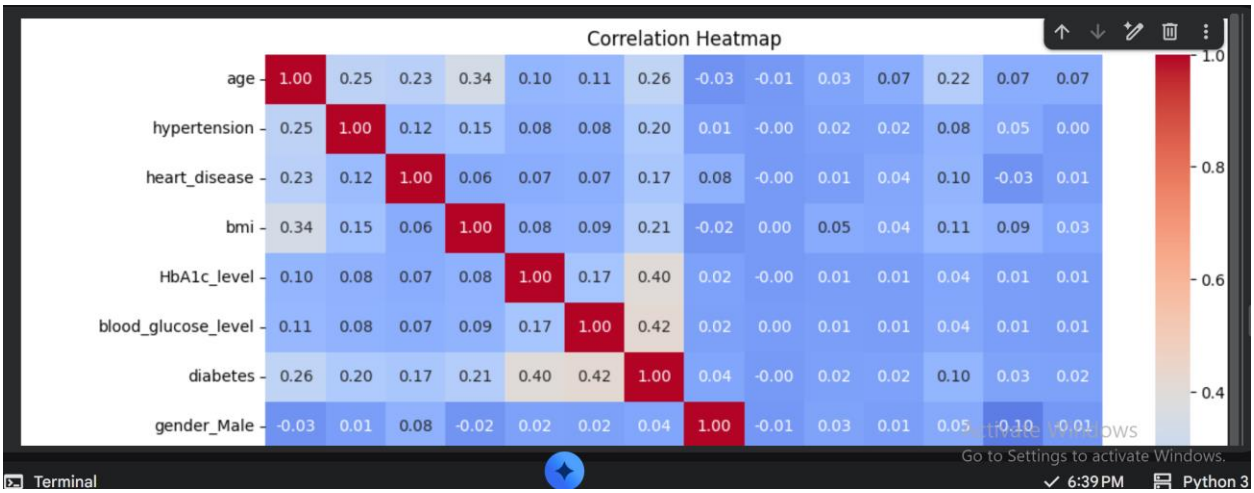
mse_multi = mean_squared_error(y_test, y_pred_multi)
rmse_multi = np.sqrt(mse_multi)
r2_multi = r2_score(y_test, y_pred_multi)

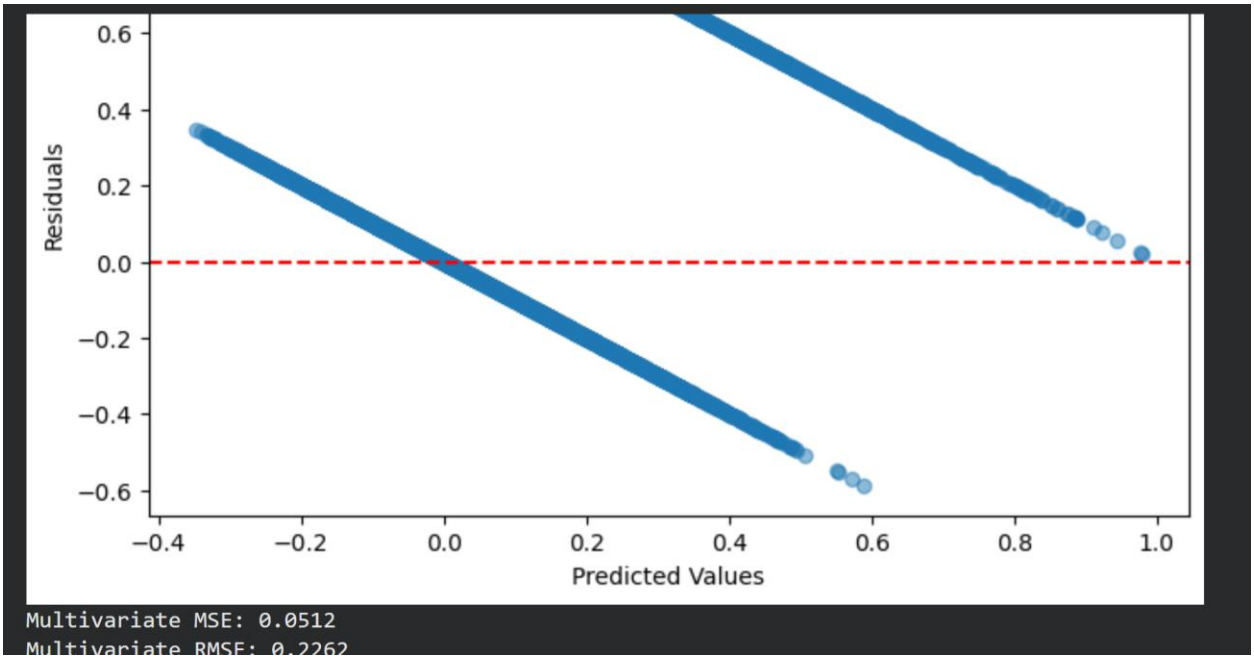
print(f"Multivariate MSE: {mse_multi:.4f}")
print(f"Multivariate RMSE: {rmse_multi:.4f}")
print(f"Multivariate R²: {r2_multi:.4f}")

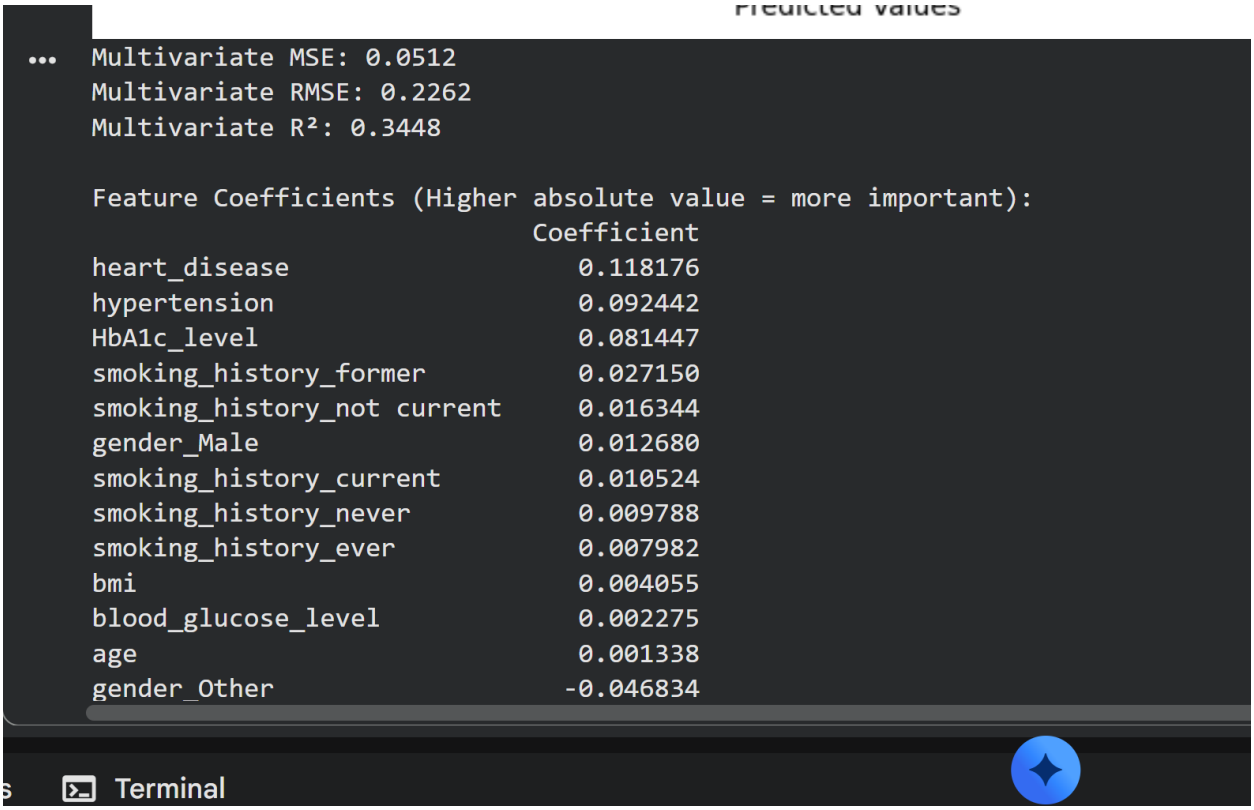
coefficients = pd.DataFrame(model_multi.coef_, X.columns, columns=['Coefficient'])
print("\nFeature Coefficients (Higher absolute value = more important):")
print(coefficients.sort_values(by='Coefficient', ascending=False))
```

Correlation Heatmap

Output:


Correlation Heatmap
age 1.00 0.25 0.23 0.34 0.10 0.11 0.26 -0.03 -0.01 0.03 0.07 0.22 0.07 0.07
hypertension 0.25 1.00 0.12 0.15 0.08 0.08 0.20 0.01 -0.00 0.02 0.02 0.08 0.05 0.00
heart_disease 0.23 0.12 1.00 0.06 0.07 0.07 0.17 0.08 -0.00 0.01 0.04 0.10 -0.03 0.01
bmi 0.34 0.15 0.06 1.00 0.08 0.09 0.21 -0.02 0.00 0.05 0.04 0.11 0.09 0.03
HbA1c_level 0.10 0.08 0.07 0.08 1.00 0.17 0.40 0.02 -0.00 0.01 0.01 0.04 0.01 0.01
blood_glucose_level 0.11 0.08 0.07 0.09 0.17 1.00 0.42 0.02 0.00 0.01 0.01 0.04 0.01 0.01
diabetes 0.26 0.20 0.17 0.21 0.40 0.42 1.00 0.04 -0.00 0.02 0.02 0.10 0.03 0.02
gender_Male -0.03 0.01 0.08 -0.02 0.02 0.02 0.04 1.00 -0.01 0.03 0.01 0.05 0.10 0.01
1.0
0.8
0.6
0.4
Activate Windows
Go to Settings to activate Windows.
Terminal
6:39 PM
Python 3


Residuals
Predicted Values
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Multivariate MSE: 0.0512
Multivariate RMSE: 0.2262



Answer:

Based on the Feature Coefficients table, the independent variables that contribute the most to predicting diabetes are heart_disease (0.118), hypertension (0.092), and HbA1c_level (0.081). Interestingly, BMI contributes very little in this multivariate model (0.004), showing that other health factors are much stronger predictors.

Task 3:

Answer:

The Multivariate Linear Regression model performs significantly better than the Simple Linear Regression model. The R^2 score increased from 0.0488 to 0.3448, and the Mean Squared Error decreased. This proves that predicting diabetes requires looking at

multiple health factors (like HbA1c level, heart disease, and blood pressure) rather than just BMI alone.